Regional Quantitative Problem-Solving Circle

Module 3: Number Theory & Modular Congruence (Weeks 5–6)

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1 Axiomatic & Theoretical Foundation

Number theory at the Olympiad level shifts focus from absolute magnitudes to the structural properties of divisibility and algebraic rings. We analyze integers not by their size, but by their residue classes and greatest common divisors.

Definition 1 (Divisibility & Modular Congruence)**.** Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say a divides b, written $a \mid b$, if there exists an integer k such that $b = ak$. Furthermore, two integers x and y are congruent modulo n, denoted $x \equiv y \pmod{n}$, if $n \mid (x - y)$.

Theorem 1 (Ring Properties of $\mathbb{Z}/n\mathbb{Z}$)**.** *The set of residue classes modulo n forms a commutative ring. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then:*

1. $a + c \equiv b + d \pmod{n}$
2. $ac \equiv bd \pmod{n}$
3. $a^k \equiv b^k \pmod{n}$ *for any integer* $k \geq 0$.

2 The Core Invariant / State Function

The foundational invariant for analyzing Diophantine divisibility is the Euclidean reduction rule.

Theorem 2 (Euclidean GCD Invariant)**.** *For any integers A, B (not both zero) and any integer k, the greatest common divisor remains strictly invariant under linear translation:*

$$\gcd(A, B) = \gcd(A, B - kA)$$

Core Invariant Framework: If $A \mid B$, then A must also divide $B - kA$ for any integer k. By selecting k strategically as an algebraic or polynomial term, we systematically eliminate higher-degree variables, collapsing the constraint to a constant numerical invariant.

3 Lemma & Canonical Proof

We demonstrate this via a classic high-degree polynomial divisibility problem, showing how Euclidean reduction forces structural collapse.

Lemma 1 (Polynomial Algebraic Divisibility)**.** *Find the maximal positive integer n such that* $(n + 10) \mid (n^3 + 100)$.

Proof. We are given the condition that $(n + 10) \mid (n^3 + 100)$.

To resolve this without empirical trial and error, we employ the Euclidean GCD invariant. We know that $(n + 10)$ divides any multiple of itself. We use synthetic substitution to construct a multiple of $(n + 10)$ that cancels the n^3 leading term. Multiplying by $(n^2 - 10n + 100)$:

$$(n + 10)(n^2 - 10n + 100) = n^3 + 1000$$

Because $(n+10)$ strictly divides $(n+10)(n^2-10n+100)$, the fundamental invariant of divisibility dictates that if $(n+10)$ divides n^3+100, it must also divide the algebraic difference between these two expressions:

$$(n+10) \mid \Big((n^3+1000)-(n^3+100)\Big)$$

Simplifying the expression on the right yields an integer constant, completely eliminating the variable n:

$$(n+10) \mid 900$$

This states that $n+10$ must be a positive divisor of 900. To maximize the positive integer n, we must select the largest possible positive divisor of 900.

The maximal divisor of 900 is 900 itself. Setting up the optimization equality:

$$n+10=900 \implies n=890$$

Thus, the maximal positive integer satisfying the divisibility condition is $n=890$. □

4 Seminar Heuristics & Socratic Framework

During the whiteboard defense phase, challenge students on the following diagnostic cues and pitfalls:

- **Diagnostic Cue — "When to apply?":** Whenever a problem asks for integer solutions where a polynomial $P(n)$ divides $Q(n)$, mandate the use of Euclidean reduction or polynomial division to isolate a numerical remainder.
- **Pitfall — Modulo Misunderstanding:** Students frequently attempt to factor n^3+100 directly, failing because it lacks rational roots. **Socratic Prompt:** *"Can we reframe* $(n+10) \mid (n^3+100)$ *as a modular congruence? What is* $n \pmod{n+10}$*? Substitute* $n \equiv -10$ *directly into the polynomial."*
- **Pitfall — Negative Divisors:** In problems asking for all integers $n \in \mathbb{Z}$, students routinely forget that negative divisors exist. **Socratic Prompt:** *"Does* $a \mid b$ *imply* $a \leq b$*? What happens when the divisor is negative?"*